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Automotive Communication Systems Lab

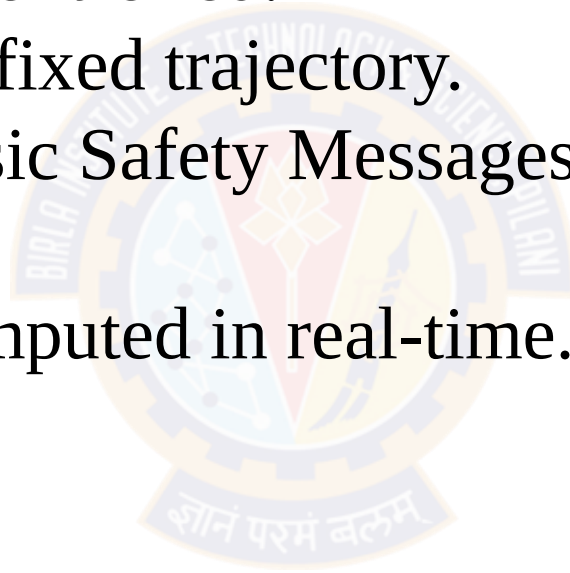
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Experiment 3 - Objectives

- ❑ To simulate Vehicle-to-Vehicle (V2V) communication using MATLAB.
- ❑ To understand Basic Safety Messages (BSM), Time-To-Collision (TTC), and Forward Collision Warning (FCW).
- ❑ To develop skills in using Automated Driving Toolbox.

Vehicle-to-vehicle (V2V) Communication

- ❑ Two vehicles: Ego Vehicle and Lead Vehicle.
- ❑ Ego Vehicle speed is controlled.
- ❑ Lead Vehicle follows fixed trajectory.
- ❑ Vehicles exchange Basic Safety Messages (BSMs).
- ❑ TTC and FCW are computed in real-time.



Scenario creation

```
scenario = drivingScenario('SampleTime',0.1,'StopTime',12);  
road(scenario,[0 0;250 0]);
```



Vehicle Definition

```
lead = vehicle(scenario,'ClassID',1,'Position',[0 0 0]);  
ego = vehicle(scenario,'ClassID',1,'Position',[-30 0 0]);
```



Trajectory Definition

`smoothTrajectory(lead,[0 0;250 0],20);`



Basic safety messages (BSM) Generation

leadBSM = generateBSM(lead,time);

egoBSM = generateBSM(ego,time);



Relative position and speed Implementation

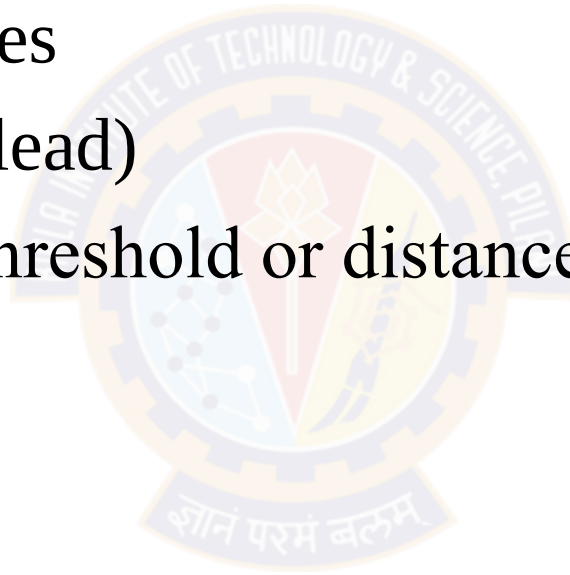
```
dx = leadBSM.Position(1) - egoBSM.Position(1);  
relSpeed = egoSpeed - leadBSM.Speed;
```



Time-to-collision (TTC) calculation

Formula: $TTC = \Delta x / v_{rel}$

- Δx : Distance between vehicles
- v_{rel} : Relative speed (ego - lead)
- FCW is triggered if $TTC < \text{threshold}$ or $\text{distance} < \text{minimum safe distance}$



FCW logic

```
if  $TTC < ttcThreshold$  ||  $dx < minDistance$   
    fcwActive = true;  
end
```



Control logic implementation

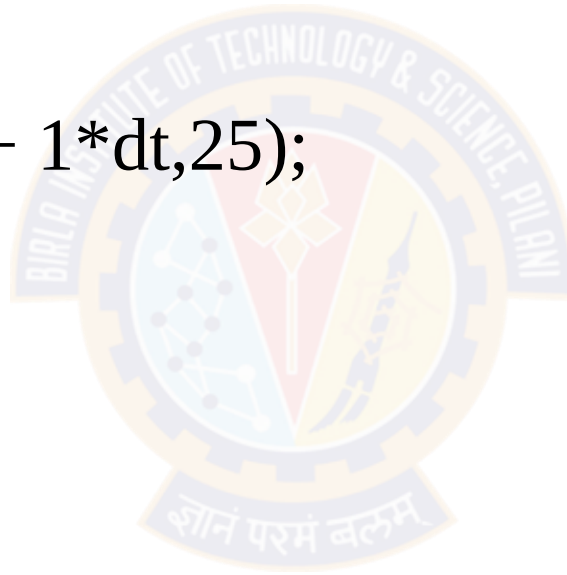
if fcwActive

 egoSpeed = max(egoSpeed - 3*dt,0);

else

 egoSpeed = min(egoSpeed + 1*dt,25);

end



MATLAB script

```
%% V2V Communication Simulation in MATLAB
% Automotive Networking and Communication Lab
% Author: [Dr.P.Priyalatha]
% Date: [January 17, 2026]
clc; clear; close all;
dt = 0.1;
T = 20;
time = 0:dt:T;
TTC_threshold = 2.5;
minDistance = 10;
lead = [0 20];
ego = [-40 25];
ego_log = []; lead_log = []; alert_log = [];
```



MATLAB script

```
for t = time
    leadBSM.Position = lead(1);
    leadBSM.Speed    = lead(2);
    egoBSM.Position  = ego(1);
    egoBSM.Speed     = ego(2);
    dx = leadBSM.Position - egoBSM.Position;
    relSpeed = egoBSM.Speed - leadBSM.Speed;
    fcwActive = false;
    if dx > 0 && relSpeed > 0
        TTC = dx / relSpeed;
        if TTC < TTC_threshold || dx < minDistance
            fcwActive = true;
            fprintf(['%.1fs] ALERT! TTC=%.2f s, dx=%.1f m\n', t, TTC, dx);
            alert_log(end+1,:) = [t TTC dx];
        end
    end
end
```



MATLAB script

```
if fcwActive
```

```
    ego(2) = max(ego(2) - 2*dt, 0);
```

```
else
```

```
    ego(2) = min(ego(2) + 0.5*dt, 30);
```

```
end
```

```
lead(1) = lead(1) + lead(2)*dt;
```

```
ego(1) = ego(1) + ego(2)*dt;
```

```
lead_log(end+1,:) = [t lead];
```

```
ego_log(end+1,:) = [t ego];
```

```
end
```



MATLAB script

```
figure;  
plot(lead_log(:,2), zeros(size(lead_log,1),1), 'r-', 'LineWidth', 2); hold on;  
plot(ego_log(:,2), zeros(size(ego_log,1),1), 'b-', 'LineWidth', 2);  
xlabel('Time (s)'); ylabel('Vehicle Lane');  
legend('Lead Vehicle', 'Ego Vehicle');  
title('V2V Communication - Vehicle Motion');
```

```
figure;  
plot(lead_log(:,1), lead_log(:,2), 'r--', 'LineWidth', 2); hold on;  
plot(ego_log(:,1), ego_log(:,2), 'b-', 'LineWidth', 2);  
xlabel('Time (s)'); ylabel('Speed (m/s)');  
legend('Lead Vehicle', 'Ego Vehicle');  
title('Vehicle Speeds');
```

```
figure;  
dist = lead_log(:,2) - ego_log(:,2);  
plot(time(1:length(dist)), dist, 'k', 'LineWidth', 2);  
xlabel('Time (s)'); ylabel('Distance Gap (m)');  
title('Inter-Vehicle Distance');
```





Thank You